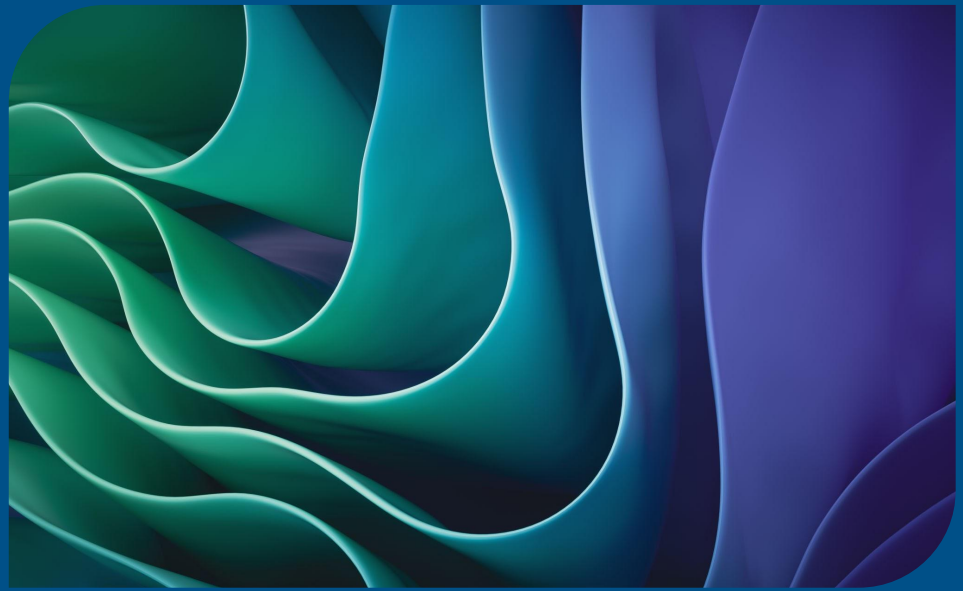
CS 6140

Investigating Double Descent

Multiple Regression Analysis on the Friedman Datasets



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1. Defining the Problem
2. Background
3. Datasets
4. Methods
5. Timeline

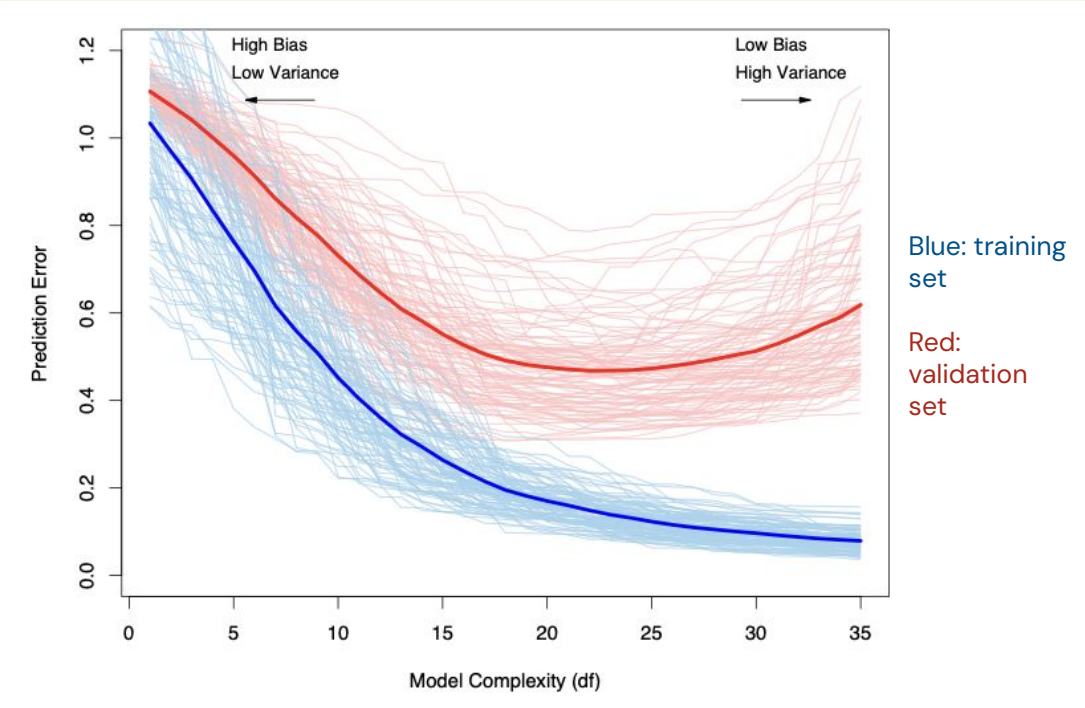
Defining the Problem

1. Classical learning theory predicts a bias-variance tradeoff as model complexity increases.
2. Recent work shows a different pattern called double descent.
3. Double descent is well-documented in neural networks, but less understood in simpler regression models.

Traditional Bias-Variance Tradeoff

As model complexity increases:

- Bias decreases.
- Variance increases.



Blue: training set

Red: validation set

Prof. Olga Vitek

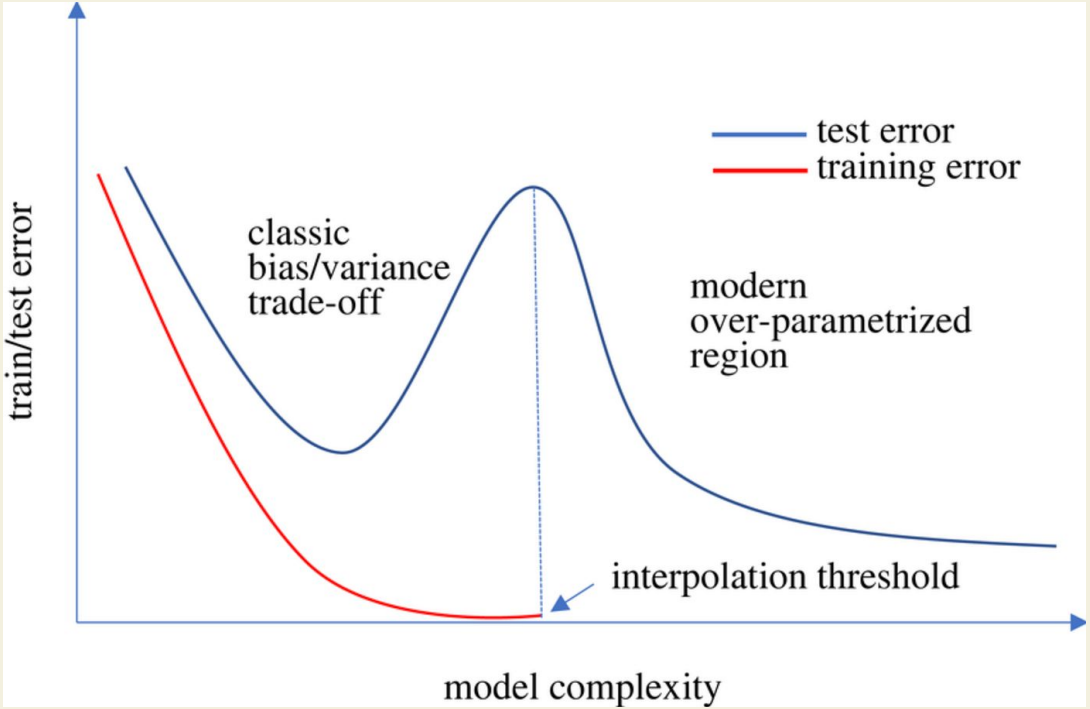
Double Descent

Recent work suggests a different behavior.

- Existence of an interpolation threshold.
- With further increases in complexity, test error can decrease again.

Models studied:

1. Polynomial Regression
2. Random Feature Regression
3. Kernel Ridge Regression
4. Neural Network



Medium

Datasets

We use synthetic datasets to ensure controlled experiments.

Properties:

- 10 predictor variables.
- Nonlinear signal structure.
- Additive noise.

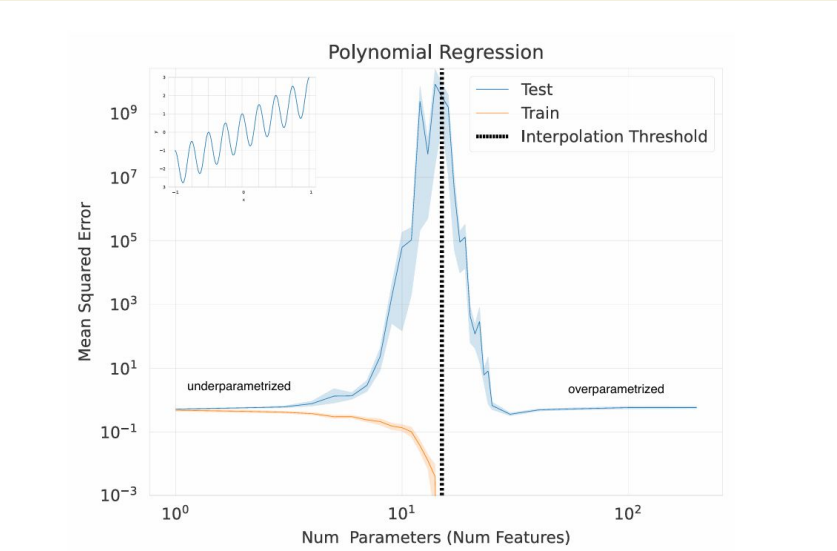
01	Friedman #1: Baseline nonlinear regression benchmark	$f(\mathbf{x}) = 10sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + 10x_4 + 5x_5 + N(0, \sigma)$
02	Friedman #2: Alternative nonlinear structure to test robustness.	$f(\mathbf{x}) = (x^2_0 + (x_1x_2 - x_1x_3)^{-1})^2)^{0.5} + N(0, \sigma)$
03	Friedman #3: Additional nonlinear formulation with different signal behavior.	$f(\mathbf{x}) = \arctan(x_1x_2 - (x_1x_3)^{-1}) / x_0) + N(0, \sigma)$

Goal: Observe training and test error behavior as model complexity increases.

Experimental Setup:

- Train/test split.
- Model complexity varied across a wide range.

1 | Polynomial Regression



- Extension of linear regression where we first transform the input features into polynomial combinations and then fit a straight line through that transformed space.

- Polynomial regression makes it easy to visualize double descent by simply increasing the degree to control model complexity.

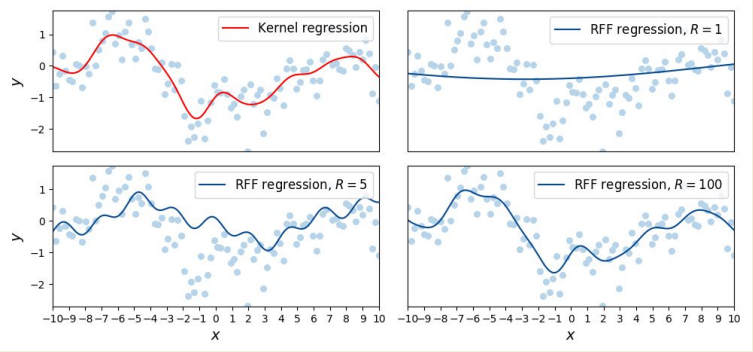
- Arxiv paper: Double Descent Demystified: Identifying, Interpreting & Ablating the Sources of a Deep Learning Puzzle

2 | Random Feature Regression (RFR)

- Faster than kernel regression using a random finite set of features
- Approximates kernel method through the inner product of chosen features

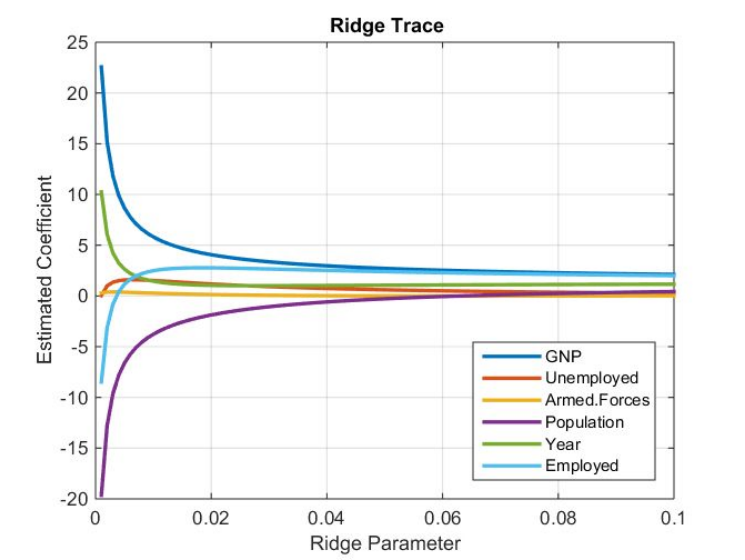
$$k(x,y) \approx \langle \mathcal{z}(x), \mathcal{z}(y) \rangle, \ \mathcal{z} : \mathbb{R}^d \longrightarrow \mathbb{R}^D$$

- Techniques such as Random Fourier Features (RFF) approximate the kernel through Fourier transforms



The size of the random features set affect the trade-off between bias and variance.

3 | Kernel Ridge Regression (KRR)



- KRR is a generalization of Linear Regression
  - Non-linear in data but linear in params
- Analyzes data that suffers from multicollinearity
- Adds L_2 penalty to the loss function Linear Regression.
- Arxiv paper: *Changing the Kernel During Training Leads to Double Descent in Kernel Regression*

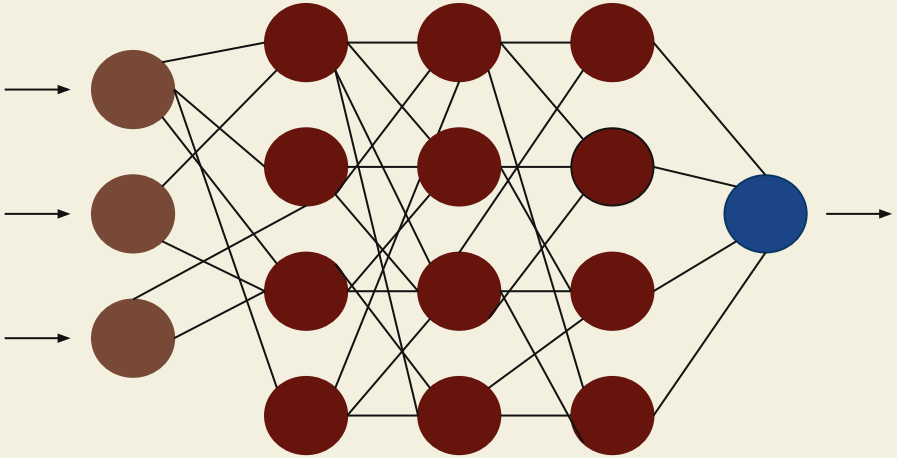
4 | Neural Network

Double descent has been extensively studied in deep learning systems.

Model characteristics:

- Fully connected feedforward network.
- ReLU activation functions.
- Trained using gradient descent.

Control model complexity with varying network width (i.e. number of hidden units).



Serves as a baseline to benchmark double descent behavior; simple vs. highly parameterized models.

Tentative Timeline

Week of 3/8/2026

Project Setup & Experimental Design

Deliverables: shared data generation script, experimental protocol, define complexity sweeps, repo structure.

Week of 3/15/2026

Baselines

Deliverables: working implementations, initial train/test error results.

Week of 3/22/2026

Double Descent Curves

Deliverables: full complexity sweeps for each models, train/test error across complexity levels, identify interpolation threshold.

Week of 3/29/2026

Validation and Comparisons

Deliverables: clean plots, cross-model comparisons.

Week of 4/5/2026

Final Analysis and Project Completion

Deliverables: final analysis and project completion.

Thank you.

References

- *Multivariate Adaptive Regression Splines, Jerome H. Friedman - The Annals of Statistics, Vol. 19, No. 1. (Mar., 1991), pp. 1-67.*
- *Changing the Kernel During Training Leads to Double Descent in Kernel Regression, Oskar Allerbo (May, 2025)*
- *Double Descent Demystified: Identifying, Interpreting & Ablating the Sources of a Deep Learning Puzzle, (March, 2023)*

Method Comparisons

M1

Polynomial Regression

Simple, parametric, interpretable.

M2

Random Feature Regression

Introduces nonlinearity with randomized features.

M3

Kernel Ridge Regression

Flexible, non-parametric approach with regularization.

M4

Neural Network

Parameterized, canonical example of double descent.